

From Observer Consensus to Standard Physics: Finite Reconstruction of Quantum Probability, Spacetime, Einstein Dynamics, and Gauge Structure

Bernhard Mueller^{1*} Alexander Osika² Mario Ponder Kai Xue Ben Cassie
Peter Nguyen Maarten Antonie Visser Kale Arnav Anirudha
David Matscheko Jonathan Hill

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Author affiliations: ¹Pragma Inc., Washington, USA; ²SNRGY Inc., Gothenburg, Sweden.

Abstract

Can operational consistency among finite observers reconstruct the mathematical structures that quantum theory, spacetime geometry, and gauge physics ordinarily postulate separately? Observer Patch Holography (OPH) studies this question on a finite architecture: observer patches with local state, boundaries, durable records, authorized overlaps, and repair maps, governed by three axioms fixing the architecture, the naturality of shared meaning, and conditional maximum randomness of everything agreement leaves free. We present the strongest results of the program in a single chain, with every result labeled by its evidence class: proved finite theorem, machine-certified finite witness, conditional reconstruction theorem, or open physical producer. Terminating confluent repair yields protected public records; the completed central record surface carries Born probabilities, Lüders conditioning, and the Tsirelson bound as finite theorems. The oriented spherical support carries $\text{Conf}^+(S^2) \cong \text{SO}^+(3, 1)$ with a three-dimensional space of observer frames. A typed composition theorem assembles modular flow, null translations, stress tomography, and entropy stationarity into the Einstein relation, with every physical antecedent displayed. Deterministic simulations measure a held-out Lorentzian event form of inertia $(1, 3)$ across a support-adjusted scaling path with a same-size control that degrades to $(2, 2)$. Twelve-port icosahedral incidence with target-blind readback yields the coefficient algebra $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$, the global form $S(U(3) \times U(2)) \cong (\text{SU}(3) \times \text{SU}(2) \times \text{U}(1))/\mathbb{Z}_6$, and a fifteen-state one-generation representation witness with exact anomaly arithmetic. Interval certificates, a positive-chamber Koide theorem whose conditional 72-eV tau enclosure carries a registered falsification protocol, and finite de Sitter capacity identities complete the quantitative surface. A formal library exceeding one thousand machine-checked declarations, executable receipts, and adversarial negative controls accompany every claim. We state precisely which attachments remain open, why the separation of evidence classes is itself a methodological contribution, and how each conditional component can fail.

Keywords: foundations of physics; observers; reconstruction theorems; quantum probability; emergent spacetime; gauge structure; formal verification; falsifiability.

*Corresponding authors: Bernhard Mueller, bernhard@floatingpragma.ai; Alexander Osika, alex@snrgystudios.com.

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1 Introduction and claim boundary

Physical theory ordinarily postulates its deepest structures separately. Quantum theory posits a Hilbert space and the Born rule; relativity posits a Lorentzian manifold; the Standard Model posits a gauge group and matter representations. Reconstruction programs ask whether some of these postulates are theorems of something smaller. This paper presents one such program, Observer Patch Holography (OPH). Its primitive is a finite society of observers that keep records, compare them where they overlap, and repair disagreement.

The central question is operational:

Can operational consistency among finite observers reconstruct the mathematical structures ordinarily postulated separately in quantum theory, spacetime geometry, and gauge physics?

Our answer is a structured partial yes, and the structure is the point. Some reconstructions are finite theorems with machine-checked proofs. Some are exact computations with executable certificates. Some are conditional compositions whose physical antecedents are stated as named premises. Some require physical producers that are open obligations. This paper never mixes those classes. Every result below carries one of four labels:

T	proved finite theorem (formal or exact-arithmetic proof)
W	machine-certified finite witness (executable certificate, controls)
C	conditional reconstruction theorem (exact, with named premises)
O	open physical producer (required attachment, no construction on record)

The discipline matters because reconstruction programs fail most often by silently promoting an interpretation into a theorem. We regard the maintained separation, enforced by fail-closed machine-readable receipts, as a contribution of the program independent of any single result.

Boundary 1.1. This paper claims no completed derivation of particle masses, no source-only determination of the fine-structure constant, no completed cosmology, and no claim that formal verification establishes physical truth. Quantitative surfaces that consume measured values are labeled diagnostics and are never counted as predictions. Section 12 lists the decisive open gates.

1.1 The three axioms

A1 (oriented twelve-port observer screen). There is an observer patch net on an oriented spherical screen: at every finite resolution each local carrier exposes twelve primitive boundary ports forming the vertices of an oriented icosahedral boundary packet with 30 edges and 20 faces; carriers join through typed seams with coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints.

A2 (observer agreement). Public meaning is natural: the interpretation map commutes with every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map on accepted public data.

A3 (conditional maximum randomness). The realized state is the information projection of an exact reference family onto the convex set of local state families compatible with the finite observer-visible constraints; the observer cover is state-determining on that feasible set with strictly positive exact weights.

None of the axioms contains a gauge group, a particle list, a spacetime dimension, or a recovery law. Everything below is an attempt to earn those structures, and the paper reports where the attempt succeeds, where it is conditional, and where it stops.

1.2 The theorem chain

The results form one connected chain:

finite carriers and records $\longrightarrow$ overlap consistency $\longrightarrow$ confluent fixed-point repair
 $\longrightarrow$ observable event algebra $\longrightarrow$ quantum probability $\longrightarrow$ observer and event geometry
 $\longrightarrow$ local Lorentzian structure $\longrightarrow$ Einstein dynamics
 $\longrightarrow$ finite gauge and matter algebra.

Sections 2 through 7 follow this chain. Section 11 describes the verification stack. Section 12 states the falsification architecture and the open obligations. Section 13 compares the program with neighboring approaches.

2 The primitive finite architecture

Definition 2.1 (Observer patch). *An observer patch is a finite object with local state, an observable boundary map, durable records, readback, and a set of authorized repair moves. A patch sees a fragment of the world: its own state and the boundary data of its authorized overlaps.*

Definition 2.2 (Overlap and repair). *For patches x_i, x_j with an authorized overlap $e = (i, j)$, both induced boundary records must agree for the pair to hold public data. A repair move changes local state while preserving protected readout. A configuration is a normal form when no authorized repair applies.*

The carrier realization used throughout is the twelve-port echosahedral architecture of A1: each carrier’s boundary packet is combinatorially the oriented icosahedron boundary (P, E, F, o) with $|P| = 12$, $|E| = 30$, $|F| = 20$; carriers federate through typed seam algebras with coherent triple-overlap cocycles into a nerve carrying a degree-one bridge to the oriented spherical support. This is the declared source architecture. We do not claim a uniqueness theorem selecting it among all possible carriers; where results depend on it, the dependence is displayed.

Two exact features of this architecture do heavy work later. First, the proper rotation group of the icosahedral packet is the alternating group A_5 , with sixty rotations acting on the twelve ports; the real port-coefficient space decomposes as

$$P_{12} \cong_{A_5} \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}, \quad (1)$$

an exact multiplicity-free decomposition with certified projectors (T). Second, record keeping is integer valued: a write appends +1, a retraction appends −1, and readback sums atomic events per port; a conservative repair transfers one unit across a seam whenever the oriented mismatch has magnitude at least two. With $V(N) = \sum_i N_i^2$, each such repair strictly decreases V by $2(d-1)$ for mismatch $d \geq 2$, so repair terminates (T). Divisibility of total load by twelve is a necessary consensus condition, settled constructively on the declared packet by an explicit witness schedule (W).

3 The fixed-point consensus theorem

The world, in this program, is a fixed point:

$$T(\mathfrak{U}) = \mathfrak{U}, \quad (2)$$

where T is the composite repair-and-agreement dynamics. The mathematical content is that the fixed point is well defined and schedule independent on the declared carrier class.

Theorem 3.1 (Consensus normal form; T/W). *On a terminating, frustration-free carrier with a confluent accepted repair relation, every repair schedule reaches the same quotient normal form. On the reference twelve-port carrier, exact code exhausts all six declared overlap schedules and returns the identical normal form; termination follows from the exact quadratic descent of Section 2.*

Boundary 3.2. Confluence is part of the carrier contract, not a free consequence of locality: arbitrary local repair does not imply it, and countermodels with non-confluent repair are retained in the formal library. The theorem is finite; no continuum limit is asserted.

Three structural properties accompany the theorem on the declared class: gauge compatibility (repair commutes with local relabeling), idempotence of the completed dynamics, and refinement compatibility where proved, each with its formal statement and, where the hypothesis fails, its countermodel. The value of the fixed point is not metaphysical: it is the object on which records become public, and every later structure is read off it.

The architecture sits close to distributed agreement in the tradition of Lamport, Shostak, and Pease [8]: patches play the role of protocol nodes, overlap repair of a quorum vote, and the completed fixed point of the decided state. A component paper of the stack proves safety and liveness for a Byzantine-tolerant reading of repair under explicit quorum and partial-synchrony assumptions; the impossibility boundary of Fischer, Lynch, and Paterson [10] is the reason those assumptions are named and are stated explicitly.

4 Observable normal forms and finite quantum probability

Once a compare, write, and verify slice has completed, its accessible events generate a finite central record algebra. The following identities are finite theorems on that surface, developed in an audited event-algebra library of more than one hundred declarations (T).

Theorem 4.1 (Finite quantum record arithmetic; T). *Let P_E be the projector of a completed public event E and ρ the normalized state of the completed record surface. Then the probability and update rules*

$$\Pr(E) = \text{Tr}(\rho P_E), \quad \rho | E = \frac{P_E \rho P_E}{\text{Tr}(\rho P_E)} \quad (3)$$

hold, and for the finite bipartite correlation surface the Clauser–Horne–Shimony–Holt parameter obeys

$$|S_{\text{CHSH}}| \leq 2\sqrt{2}. \quad (4)$$

The first identity is the Born rule on this finite surface; the second is Lüders conditioning; the third is the Tsirelson bound [7] for the Clauser–Horne–Shimony–Holt combination [6] at fixed cutoff. Compatibility structure for commuting events follows the standard finite operational pattern within the same library.

Boundary 4.2. This is finite operational quantum structure derived from public observability. It is not a continuum interacting quantum field theory, and no claim of that kind is made anywhere in this paper. Its role in the chain is specific: the probability and update rules that later physical branches require arise from the consensus architecture.

5 From observer frames to events and spacetime

5.1 Exact kinematics of the spherical support

The oriented conformal spherical support carries

$$\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1), \quad H^3 \cong \frac{\text{SO}^+(3, 1)}{\text{SO}(3)}, \quad \dim H^3 = 3. \quad (5)$$

These are classical isomorphisms of conformal geometry, consumed as exact mathematics. The Lorentz group and the three-dimensional space of unit timelike observer frames are exact kinematic consequences of the support branch.

Boundary 5.1. Observer-frame geometry and populated event geometry are distinct objects, and the program keeps them distinct. A populated four-dimensional event base requires record separation, local charts, an open-image condition, affine transition data, a quadratic cone, and causal reachability; these are producer obligations, not corollaries of (5). The frame space H^3 and event localization are typed as separate constructions, and no inference passes from one to the other without a certificate.

5.2 A measured finite Lorentzian signature

The simulation branch measures the event form. Repair dynamics on federated carriers emits semantic events with ancestry; a declared chart combines one ancestry-depth coordinate with three spectral coordinates; a quadratic event form is fitted on a declared training half of the event pairs and evaluated on the held-out half. The signature is therefore held out from the update rule (W).

On the selected support-adjusted path with (carrier count, observer count, support width) equal to (16,384, 128, 96), (65,536, 256, 96), and (262,144, 512, 384), the held-out event form has inertia (1, 3) at every rung, with cone margins

$$-5.6, \quad -3.2, \quad -1.4 \quad (6)$$

whose magnitudes shrink along the path. The mechanism control is decisive for interpretation: at 262,144 carriers the same-size support-width-96 control has 312 cross-observer edges and inertia (2, 2), while the support-width-384 row has 1,062 edges and inertia (1, 3). The signature tracks cross-observer coupling, not the fitting machinery.

Boundary 5.2. The chart allocates four coordinates by declaration; the instrument tests signature and full rank on that chart and does not derive the number of spacetime dimensions. The negative cone margins, one Euclidean local fit in the finite source domain below, closed neighborhoods in place of open charts, and the absence of a certified refinement limit prevent promotion of any finite object in this section to a continuum Lorentzian manifold. These conditions are recorded machine-readably in the receipts (Ofor the continuum attachment).

5.3 A target-clean source inhabits the finite local domain

One deterministic capture at 16,384 carriers supplies: a finite causal complex of 2,304 events with exact acyclicity and a strict time function; a certified four-column chart, nondegenerate on the sample, whose global held-out form has inertia $(1, 3)$; six closed observer-visibility neighborhoods with exact induced affine transitions, cocycles, orientation, and time orientation; an observer-visible seam complex of 8,662 carriers and 11,816 seams whose 38 triangles are all frustrated under the declared orientation-reversing seam transport, with lift-ambiguity rank 3,117 over $\mathbb{F}_2$; typed scalar, chiral, and gauge sections with a sign-twisted local derivative whose adjoint, kinetic, covariance, gluing, refinement, and boundary identities are verified by exact integer evaluation; and a signed-graph rank theorem giving a zero twisted kernel, hence an exactly positive dimensionless spectral gap for the declared local operator (W , with the rank and positivity statements T).

One target-clean provenance graph binds the construction; an isolated rerun reproduces canonical receipt content byte for byte. The receipts record, fail closed, that this finite domain does not promote to a continuum spacetime, a physical clock, or a physical mass scale.

6 Einstein reconstruction

The gravitational branch is a composition theorem with every antecedent visible:

Theorem 6.1 (Einstein composition; C). *On one common algebra-state tower, the typed premises*

$$\text{Event} + \text{Modular} + \text{Stress} + \text{Entropy} + \text{SmallDiamond} + \text{Scale}$$

compose to

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle. \quad (7)$$

Normalized modular flow supplies the local clock; half-sided modular inclusions supply positive null translations; null tomography assembles local stress; generalized-entropy stationarity on small causal diamonds supplies the field equation.

The formal library proves the typed algebraic composition. The inventory of what stands where: the composition itself is exact (C); the finite instruments that would populate its premises on a source tower exist and are adversarially tested (W); a single source-derived physical tower satisfying all premises simultaneously, the continuum limit, and the physical identification maps are open producers (O). The signature measurement of Section 5.2 is evidence that the source dynamics generates the causal-geometric raw material the premises require; it is not a substitute for them.

This structure is the opposite of an emergence slogan: it is a contract. It states exactly which receipts a physical carrier must supply for the Einstein relation to hold on it, and the program's simulation branch measures progress against that contract.

7 Gauge structure and a one-generation matter witness

7.1 The coefficient algebra

Fix the twelve-port packet with its six antipodal axes u_i and projectors $P_i = u_i u_i^\top$. Target-blind port impulses with local recurrence readback derive the inverse response $R = -J$ with

$J = (A^3 - 4A^2 - 5A + 10I)/10$ in the adjacency algebra, and the face orientation fixes the companion three-dimensional map. Pulling the block commutator back through the resulting equivariant isomorphism yields

$$(P_{12}, [\cdot, \cdot]_{\Theta}) \cong \mathfrak{u}(3) \oplus \mathfrak{so}(3) = \mathfrak{u}(1) \oplus \mathfrak{su}(3) \oplus \mathfrak{su}(2) \quad (8)$$

as an exact finite computation: the map is injective, its image closed under commutators, the five-band noncentral, the derived algebra eleven-dimensional, the center the uniform line, and the action natural along declared refinements (T/W). The derivation consumes no target data: the readback protocol is fixed before any comparison with Standard Model structure.

7.2 Global form

Trace balancing extends the block construction to $\mathfrak{s}(\mathfrak{u}(3) \oplus \mathfrak{u}(2))$, whose connected group is

$$S(U(3) \times U(2)) \cong \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}. \quad (9)$$

The six-element kernel is exact, as is the independent lattice quotient with Smith invariants $(1, 1, 1, 1, 1, 6)$ (T). A measured transport artifact supplies the non-split deck cover with its unique central implementation of the odd-Weyl grading, and the measured finite flux-sector menu selects the $\mathbb{Z}_6$ quotient on the declared carrier (W).

7.3 One generation of matter, exactly

The trace-balanced block $V = \mathbb{C}^3 \oplus \mathbb{C}^2$ with hypercharges $(-\frac{1}{3}, \frac{1}{2})$ has the exterior branching

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L : \quad (10)$$

fifteen states carrying the Standard Model hypercharges of one generation, the expected invariant lines, cancelled gauge, mixed, and gravitational anomalies, and exactly four weak doublets, hence even Witten parity (T). An exhaustive scan of all 1,024 exterior subsets classifies the alternatives: the anomaly-free chiral survivors are exactly the unordered charge-conjugate pair of this configuration (W).

7.4 Family-band selection

Within the screen coefficient module (1), restricting to single complete faithful band objects in the independently bounded window $3 \leq N_g \leq 5$, the exact operational Laplacian costs $5 - \sqrt{5} < 6 < 5 + \sqrt{5}$ select the rank-three band uniquely, and the declared unitary simulator realizes that projector at its lowest positive generator frequency (Cfor the selection under its named interface; Wfor the realization). Tensoring with the generation table gives a conditional complex rank-45 candidate.

7.5 Massless carrier statements

The breaking pattern encoded in (9) makes three carrier statements available, each exact at its own level and each bounded by a named constraint.

The photon. On the finite one-loop electroweak stack built over the canonical action, two facts are machine checked: the longitudinal part of the photon self-energy vanishes identically on the

minimum-restored surface, including finite parts, and the photon direction annihilates every generator of the rotated neutral mass matrix, so the photon mass term is exactly zero in the finite calculation (W). Charge universality holds in the same stack through the certified combination $\delta Z_e = -\frac{1}{2}\delta Z_{AA} - \frac{s_w}{2c_w}\delta Z_{ZA}$, which is gauge independent and reproduces the census beta coefficients. The constraint: these are statements of the validated external-consumer calculation on a declared synthetic fixture; the physical photon pole verdict additionally requires the physical current attachment of the boundary above.

The gluon. The color factor $\mathfrak{su}(3)$ in (8) lies in the unbroken subalgebra of the global form: no scalar direction in the realized matter and descent structure carries color, so color symmetry is exact and the finite gluon mass term is zero (T at the algebra level). The constraint is stated with equal weight: confinement, the physical gluon spectrum, and any statement about hadrons belong to strong-coupling dynamics that the program deliberately does not simulate; the corresponding physical row closes as not evaluable under the declared no-QCD resource boundary and carries no physical spectrum claim.

The graviton. The Einstein branch of Theorem 6.1 carries a massless spin-two carrier exactly as far as the composition carries the field equation: linearizing the composed relation about a vacuum yields the standard massless graviton kinematics, conditional on the same premise vector, and inheriting its openness (C). No independent graviton claim is made beyond the composition.

These three statements illustrate the program’s division of labor at its best: exact finite masslessness where the algebra forces it, machine-checked cancellation where a calculation certifies it, and explicit refusal where dynamics outruns the declared resources.

Boundary 7.1. The following are open producers (O), stated with the same prominence as the results: laboratory current attachment; identity of the response current with the independently reconstructed categorical current; physical matter identification and matter-pole structure; exclusion of extra light sectors; physical seam-action selection; a certified bridge between the twelve-port spin packet and the local operator domain; three physical families as a source-selected fact; Yukawa couplings, mixing, and masses. The program’s own identifiability audits prove that several of these cannot be extracted from the registered source interface, so they are listed as producers.

8 Quantitative closure surfaces

Three quantitative surfaces show that the architecture engages measured physics. Each is stated with its epistemic class; none is a source-only prediction of a dimensionful constant.

8.1 Interval-certified fixed points

The local screen cell is asked to reproduce its readable electromagnetic width through the closure $P = \varphi + \sqrt{\pi}/A_T(P)$ with $\varphi = (1 + \sqrt{5})/2$. For each declared map, outward-rounded interval certificates prove existence and local uniqueness of the root and exclude a second root across the analytic domain (W). The gauge-width map’s certified root sits at relative separation 2.5×10^{-6} from the measured inverse fine-structure constant [11].

Boundary 8.1. The physical interpretation requires a source-derived same-scheme hadronic transport that does not exist; the comparison is a diagnostic. The certified roots do not constitute an independent determination of the fine-structure constant, and the program’s fail-closed ledgers prevent any such promotion.

8.2 The positive-chamber Koide theorem and a registered falsification window

The charged-lepton combination Q below is the one identified by Koide [9]; the theorem locates its value in an exact positivity chamber, not from a fit.

Theorem 8.2 (Positive-chamber circulant identity; T). *For the Hermitian three-cycle response $C = aI + bR + \bar{b}R^2$ with $R^3 = I$ and nonnegative eigenvalues read as square-root masses,*

$$Q = \frac{\sum_i m_i}{(\sum_i \sqrt{m_i})^2} = \frac{1}{3} + \frac{2}{3} \left(\frac{|b|}{a} \right)^2, \quad \text{so} \quad Q = \frac{2}{3} \iff \frac{|b|}{a} = \frac{1}{\sqrt{2}}. \quad (11)$$

Equal rank-two event blocks in the finite tracial packet supply the balance exactly.

Under the balance and mass-ordering premises, the measured electron and muon masses determine the tau mass inside an outward-rounded 72-eV enclosure, [1776.968991, 1776.969063] MeV, sitting 0.43σ from the measured tau mass [12]. The premise ancestry is declared: the balance premise is target informed, so this is a conditional postdiction with a registered falsification protocol, precision floor 0.045 MeV, and a kill band fixed before comparison (Wfor the enclosure; Cfor the physical reading). A qualifying future measurement outside the window refutes the balanced-circulant premise.

8.3 Finite de Sitter capacity identities

For sector dimensions d_i with total M and probabilities p_i ,

$$S_{\text{gen}}(p, d) = \log M - D_{\text{rel}}(p \parallel d/M), \quad (12)$$

so the maximum $\log M$ is attained exactly at $p_i = d_i/M$; and under uniform depletion with transferred fraction f , every admissible transfer obeys $\Delta S_{\text{gen}}^{\text{max}} = \log(1 - f) < 0$, strictly decreasing and concave (T). In pure de Sitter space the radius cancels from the shock normalization, $\mu^2 = d - 2 = \lambda_{\ell=1}(S^{d-2})$, with the $\ell = 1$ modes exactly the isometry gauge kernel (T). Identifying capacity with horizon area and the finite operator with the gravitational shock would supply a time-advance mechanism; the identification maps are open producers (O), and a related static-patch trace proposal is excluded by the obstruction of Chen, Stanford, Tang, and Yang [13], which the finite identities above do not contradict.

9 Proof-bearing finite construction

This section records the mathematical core in more detail than the overview above. It is included so that the logical load does not reside only in a repository citation. Each result remains scoped to the finite carrier and to the hypotheses displayed with it.

9.1 Descent, local diamonds, and a unique public record

Let $q = (q_p)_{p \in P}$ be the integer readback vector on a twelve-port carrier. For an oriented seam $p \rightarrow p'$ with $q_p - q_{p'} = d \geq 2$, the elementary repair sends $(q_p, q_{p'})$ to $(q_p - 1, q_{p'} + 1)$. The quadratic record cost is $V(q) = \sum_{p \in P} q_p^2$. Direct expansion gives

$$V(q') - V(q) = (q_p - 1)^2 + (q_{p'} + 1)^2 - q_p^2 - q_{p'}^2 = -2(d - 1) < 0. \quad (13)$$

The total load is preserved. Since V is a nonnegative integer, no repair trajectory is infinite.

Lemma 9.1 (Finite termination). *Every trajectory generated by the declared conservative repair moves ends in a normal form.*

Proof. Equation (13) decreases V by at least two at each step. A strictly decreasing sequence of nonnegative integers is finite. $\square$

Termination alone does not identify the endpoint. The carrier contract therefore supplies a local diamond: whenever two accepted repairs start from one configuration, their outputs have a common repair successor that preserves the protected readout. This is the finite rewrite hypothesis used in the normal-form theorem. It has the same formal role as the local confluence assumption in Newman's lemma [14]; the exact carrier proof checks the permitted overlaps and retains failed mutations as counterexamples.

Theorem 9.2 (Protected consensus normal form; T). *On a terminating carrier satisfying the declared local diamond and protected readback clauses, every accepted repair schedule has one quotient normal form. The induced record is independent of the schedule.*

Proof. The local diamond and Lemma 9.1 give confluence by the standard induction on the sum of the two remaining descent lengths. Two terminal descendants of one state have a common descendant by confluence and are therefore equal in the quotient. Protected readback is invariant along each edge of the repair relation, so it descends to that quotient. $\square$

The theorem identifies the exact point at which private writes become public: the public object is the schedule-independent quotient record, not an arbitrary intermediate configuration. The result has a distributed-systems analogue in Byzantine agreement [8]; its assumptions are necessary because asynchronous agreement with faults has the Fischer–Lynch–Paterson obstruction [10].

9.2 Finite event algebra

The completed record is represented by a finite-dimensional $*$ -algebra $\mathcal{A} = \bigoplus_j M_{n_j}(\mathbb{C})$ with a normalized positive functional $\omega(A) = \text{Tr}(\rho A)$. A completed event is a projector $P_E = P_E^* = P_E^2$. Positivity and normalization give

$$0 \leq \text{Tr}(\rho P_E) \leq 1, \quad \sum_k \text{Tr}(\rho P_{E_k}) = 1 \quad (14)$$

for every complete orthogonal event partition (P_{E_k}) . Thus the Born probability is a theorem of the record surface once its state and event projectors have been constructed. Conditioning on an event of nonzero probability gives the unique normalized positive functional supported on the post-event corner,

$$\rho_E = \frac{P_E \rho P_E}{\text{Tr}(\rho P_E)}. \quad (15)$$

For self-adjoint contractions A_0, A_1, B_0, B_1 in two commuting record subalgebras, set $S = A_0(B_0 + B_1) + A_1(B_0 - B_1)$. The commuting-subalgebra calculation gives

$$S^2 = 4I - [A_0, A_1][B_0, B_1], \quad \|S\| \leq 2\sqrt{2}. \quad (16)$$

The last inequality follows from $\|[A_0, A_1]\|, \|[B_0, B_1]\| \leq 2$. Taking the state expectation proves the finite Tsirelson inequality. The original Bell–CHSH question and the quantum bound are due to Clauser, Horne, Shimony, and Holt [6] and Cirel'son [7]. The OPH statement concerns the constructed finite record algebra, with no assertion that this establishes an interacting continuum quantum field theory.

9.3 The geometry-to-dynamics composition

The spherical support supplies a group-theoretic kinematic datum:

$$\text{Conf}^+(S^2) \cong \text{PSL}(2, \mathbb{C}) \cong \text{SO}^+(3, 1). \quad (17)$$

Its stabilizer of a unit future timelike vector is $\text{SO}(3)$, so the observer-frame quotient has dimension $6 - 3 = 3$. This fact gives the local Lorentz group on the support; it does not construct an event manifold. The event construction carries separate finite receipts for causal order, chart rank, affine overlaps, orientation, and time orientation.

The Einstein result is a typed implication. Let $\mathcal{D}$ denote a common domain with a state, modular generator, null-translation structure, stress tomography, and generalized entropy functional. The premise contract is

$$\delta K = \delta \langle H_{\text{mod}} \rangle, \quad (\text{modular first law}), \quad (18)$$

$$\delta S_{\text{gen}} = 0, \quad (\text{small-diamond stationarity}), \quad (19)$$

$$\delta \langle H_{\text{mod}} \rangle \propto \int \xi^a \delta \langle T_{ab} \rangle d\Sigma^b, \quad (\text{stress tomography}). \quad (20)$$

With the stated normalization and scale maps, the first variation is the Einstein relation in Theorem 6.1. This architecture is close to the thermodynamic route developed by Jacobson [18, 19] and uses modular ingredients studied by Bisognano and Wichmann [15, 16] and Wiesbrock [17]. The paper's claim is the finite, typed composition of the premises. A physical tower realizing the whole premise vector remains an open producer.

9.4 Gauge algebra and anomaly arithmetic

The twelve-port coefficient space has the exact alternating-group split

$$P_{12} \cong \mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}. \quad (21)$$

Six antipodal axes give rank-one projectors $P_i = u_i u_i^\top$. The response map constructed from target-blind port impulses identifies the noncentral coefficient blocks with a trace-balanced $3 \oplus 2$ block system. The commutator is therefore

$$[\Theta(x), \Theta(y)] = \Theta([x, y]_\Theta), \quad (P_{12}, [\cdot, \cdot]_\Theta) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3). \quad (22)$$

Injectivity, closure, the eleven-dimensional derived algebra, and the one-dimensional center are finite checks of the map Θ . The global trace condition gives $S(U(3) \times U(2))$, with common kernel $\mathbb{Z}_6$. This representation-theoretic route is compared with the Doplicher–Roberts reconstruction framework [20, 21] and with the standard grand-unification embedding of Georgi and Glashow [22]; it is not an assertion that either external framework proves the OPH carrier map.

For $V = \mathbb{C}^3 \oplus \mathbb{C}^2$ with charges $(-1/3, 1/2)$, the exterior package

$$\Lambda^2 V \oplus \Lambda^4 V = Q \oplus u^c \oplus e^c \oplus d^c \oplus L \quad (23)$$

contains fifteen chiral states. The finite anomaly checks evaluate the cubic color, cubic weak, mixed gauge, and gravitational sums on this table; each vanishes, while the number of weak doublets is four and hence has even Witten parity [23]. The arithmetic selects a representation witness. Physical matter identification, Yukawa couplings, and the observed mass spectrum require additional physical maps.

10 Physics questions fixed by a physical OPH realization

The theory contains results with different antecedents. The phrase “solved if OPH holds” has a precise meaning here: if a physical carrier realizes the stated source object and the named interpretation maps, the corresponding conclusion follows without an additional phenomenological postulate. The table separates those consequences from problems that remain outside the stated result.

Physics question	Consequence of the stated realization	Boundary
Why do completed observations obey quantum probability? Why is local kinematics Lorentzian?	Public event projectors obey (14); completed-event conditioning is (15); bipartite record correlations obey (16). The spherical support has the connected Lorentz symmetry (17); the event instrument measures inertia (1, 3) on its declared finite path.	The theorem is finite and does not construct a continuum field theory. A physical continuum event manifold and refinement limit require their own receipts.
What equation governs gravity?	A common physical realization of the premise contract (20) satisfies the Einstein relation.	The common physical tower, normalization, and clock identification are open.
Why this compact gauge Lie type?	The port response bracket fixes $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ and the trace-balanced quotient fixes the $\mathbb{Z}_6$ kernel.	Laboratory-current attachment and the link to the categorical current are open.
Why one anomaly-free generation table?	The exterior package (23) gives the fifteen-state chiral table with anomaly cancellation.	It does not select physical matter fields or their masses.
Why a massless photon, gluon, and graviton description?	The finite Ward calculation cancels the photon mass term, unbroken color has no finite mass term, and linearization of the Einstein branch gives massless spin-two kinematics.	Physical poles, confinement, and a physical graviton require their respective continuum and attachment records.
What fixes horizon entropy transfer?	The finite capacity theorem gives the unique entropy maximum and the sign $\Delta S_{\text{gen}}^{\text{max}} < 0$ for a declared transfer.	Horizon-area and shock dictionaries are physical identification maps.

10.1 How gravity emerges

OPH begins with finite observer patches instead of a metric field. Each patch has local state, a boundary, protected records, readback, and repair moves. Confluent repair turns compatible overlap records into a public quotient record. The oriented support of that record construction carries the connected conformal group of the two-sphere, which is the proper Lorentz group. This supplies the kinematic arena and the local observer-frame space. A separate event producer turns record separation and causal ancestry into finite event data with chart, cone, transition, and causal-order certificates.

The dynamical step uses one common algebra-state tower. Its modular flow defines the local time generator; half-sided inclusions provide the positive-null translation structure; stress tomography connects the flow to local energy data; generalized-entropy stationarity on small causal

diamonds then gives the Einstein relation with the normalization carried by the same tower. This is the end-to-end claim: finite self-reading patches supply public records, public records supply the support and event inputs, and the completed modular-entropy package supplies gravity. The finite signature instrument measures the expected $(1, 3)$ event form on its declared path. The outstanding task is a physical construction of the common tower, its continuum refinement, and its clock and stress identifications.

Intuition. Gravity is the rule that keeps local observer accounts mutually compatible as their records are extended, compared, and repaired. The metric is the geometrical description of the stable large-scale relation among those repaired records. Modular flow tracks how a local observer reads change, and entropy stationarity fixes the gravitational response required for that reading to remain consistent.

10.2 What unifies gravity and the Standard Model

The unifying object is not a conventional grand-unified gauge group added to an independent gravitational sector. It is the observer-patch architecture itself. Its overlap, record, and repair structure feeds the geometry branch, while its twelve-port incidence, response map, and trace-balanced coefficient blocks feed the gauge branch. The same finite carrier therefore supplies both the data from which local Lorentzian geometry is assembled and the data from which the compact gauge algebra and its global quotient are calculated.

This is a unification by common construction and common evidence boundary. It does not identify gravity with a gauge boson or claim that every physical coupling follows from the finite coefficient calculation. If one physical carrier realizes both branches with their named maps, the Einstein relation, the Standard Model Lie type, the $\mathbb{Z}_6$ global form, and the one-generation anomaly-free representation package are consequences of the same observer-first source. Matter poles, scalar dynamics, family attachment, and laboratory-current identification remain separate physical questions.

Intuition. One object is doing two jobs. Its overlap structure organizes the event and gravity route, while its finite boundary response organizes the charge route. Unification means that space-time and gauge structure are two readouts of the same self-reading carrier, with neither branch inserted as an independent starting ingredient.

10.3 Why quantum measurement has a physical location

The measurement problem is addressed at the record level. An observation is not represented by an external observer selecting an outcome; it is a completed public record that survives the authorized overlap checks. The event projector, its probability, and the update of the record state are defined on that completed surface. This supplies the Born and Lüders rules for the finite construction and places the distinction between an unfinished private process and a settled public event in the repair dynamics. A physical realization would have to show that its durable records and readback maps instantiate this finite construction. The result does not settle every interpretive question about continuum quantum field theory.

Intuition. A measurement becomes objective when its record has survived every authorized comparison and no repair can change it. Probability describes the completed record surface; collapse

is the update from one public record class to the next. The construction locates observation in the physical bookkeeping that patches perform on their shared boundaries.

10.4 Why string theory can be an effective description

The screen construction has boundary degrees of freedom, seam transport, edge observables, and a regulated refinement structure. Those are the features from which a string-like edge description can arise as an effective language. In this role, string variables encode the carrier’s boundary transport and consistency conditions; their validity is assessed by whether they preserve the observer-record, current, clock, and refinement receipts. The string-vacuum analysis in the OPH stack uses these gates to classify candidate effective edge descriptions [5].

This claim does not derive a unique critical string theory, select a vacuum, or establish a fundamental worldsheet ontology. It states a direction of explanation: if the observer carrier is fundamental, a string description can be useful because it captures organized boundary transport on that carrier. The physical selection of a string-effective regime requires its own carrier and spectrum receipts.

Intuition. Strings are useful here as organized boundary motion. A patch carrier has edges, seams, and transport around closed boundary circuits. When those degrees of freedom admit a string description, the string records their collective propagation. The observer carrier remains the underlying object whose records and repair rules decide whether that description is admissible.

10.5 What becomes of cosmological and horizon questions

The capacity branch identifies a precise finite question: how much public record capacity is available across a declared screen partition, and how does its entropy respond to a transfer? The exact answer is the unique relative-entropy maximum and the negative transfer sign in the table above. With a physical horizon-area dictionary and a shock-operator dictionary, this becomes a statement about de Sitter entropy transfer and a time-advance mechanism. The finite theorem does not provide those dictionaries. It therefore fixes the mathematical response law while leaving its physical attachment open.

Intuition. The screen has limited room for jointly readable records. Moving a share of that room changes the entropy available to the remaining observer horizon. The finite theorem fixes the sign and curvature of this accounting. A physical horizon interpretation would turn that accounting rule into a gravitational response law.

11 Machine verification and reproducibility

Every labeled result maps to an artifact class:

- a sorry-free Lean library exceeding one thousand theorem and lemma declarations, whose count deliberately includes premise boundaries and countermodels, because exposing assumptions is the purpose of the formalization;
- exact-arithmetic code receipts: integer and rational computations, interval certificates with outward rounding, and $\mathbb{F}_2$ rank computations, each emitting a canonical hashed payload;

- simulation receipts pinning source commit, configuration, seeds, grids, tolerances, and output hashes, with deterministic reserialization distinguished from fresh source replay;
- fail-closed verifiers that recompute verdicts from clause vectors and refuse receipts whose stored verdict disagrees, so a truth flag can never be asserted by a caller;
- adversarial negative controls: every certificate ships with mutations that must be detected, and a control that cannot fail is treated as a defect of the certificate, not as support;
- no-target-ancestry rules: machine-readable ledgers classify every quantitative row as source result, conditional composition, diagnostic, or rejected candidate, and forbid promotion by prose, issue state, or numerical proximity.

The repository, receipts, schemas, and rebuild instructions are public [2, 3]. The verification stack does not prove OPH physically true; it proves that the stated theorems hold and that the stated boundaries are the ones actually enforced.

11.1 Artifact map

Each labeled result of this paper resolves to named artifacts. The formal library counts 1,008 public theorem and lemma declarations at the pinned release, with a continuous-integration floor of 1,000.

Result	Principal artifacts
Consensus normal form; schedule exhaustion	<code>Lean/ObservableNormalForms/...</code> / <code>ObserverConfluence.lean</code> ; exact schedule code receipts
Event algebra, Born, Lüders, Tsirelson	<code>Lean/EventAlgebra/</code> (<code>Basic</code> , <code>Lueders</code> , <code>Tsirelson</code> , <code>ExpectationBound</code> , partition and state modules); per-module axiom audits
Port action, commutant, coefficient algebra	<code>Lean/Screen/A5PortAction.lean</code> , <code>A5Commutant.lean</code> , <code>A5OPH.lean</code> , <code>A5IncidenceResponse.lean</code>
Global form and $\mathbb{Z}_6$ kernel	<code>Lean/Screen/Z6Exact.lean</code> ; exterior scan in <code>ExteriorSelection.lean</code>
Family-band selection	<code>Lean/Screen/A5FamilyBand.lean</code> ; band-attachment certificate and pole-residue artifact in <code>code/a5_closure/</code>
Equal state weights (A3 forcing)	<code>Lean/Screen/EqualStateWeights.lean</code>
Einstein composition	<code>Lean/ObserverPatchHolography/EinsteinBranch</code>
Signature ladder and control	<code>evidence/einstein_convergence/</code> (four-row manifest, hashed arrays; regeneration script in the simulator repository)
Finite local source domain	staged receipts under <code>data/local_domain/</code> and the bundle verifier under <code>oph_fpe/local_domain/</code> in the simulator repository
Interval fixed points; lepton intervals; Koide window	interval certificates and custody records under <code>code/</code> and <code>claims/</code> [4]
Photon cancellation and charge universality	one-loop stack receipts under <code>code/particles/calibration/wz_upstream_completion/</code>

12 Empirical meaning, falsifiability, and open obligations

The program maintains a preregistration architecture: custody-bound target definitions, kill bands, precision floors, and decision rules fixed before comparison, with falsifications retained at the same prominence as confirmations [4]. Standing registered items include the conditional tau window above and a ringdown comb stance re-tested against each sufficiently loud gravitational-wave ring-down. Candidate hypotheses that failed their predeclared comparisons are retained in a negative ledger and contribute nothing to the case made here.

The program’s empirical case rests on three decisive gates:

1. **Physical realization of the common refinement tower.** The Einstein composition of Theorem 6.1 needs one source-derived tower satisfying its premises jointly; the signature instrument and its controls measure progress toward exactly this object.
2. **A source-only current and continuum producer.** The gauge and matter results of Section 7 await laboratory current attachment and a continuum operator limit; identifiability audits delimit which attachments the registered source interface cannot supply. This gate requires source-visible observables beyond that interface.
3. **One preregistered quantitative forecast not used in architecture selection.** A sealed machine-readable prediction contract fixes the comparison custody. The first source-visible candidate is scored once; a complete negative inventory is an acceptable published outcome.

Each gate can fail, and the second one can fail provably. The program treats a rigorous non-identifiability theorem as a result of equal standing with a positive construction, because it tells the theory where additional observable structure must come from.

13 Discussion and comparison

The components of OPH have recognizable neighbors. Operational reconstructions of quantum theory derive Hilbert-space structure from information-theoretic postulates; OPH’s Section 4 differs by deriving the finite record calculus from a consensus fixed point rather than from axioms about state spaces. Algebraic and modular approaches to quantum gravity use Tomita–Takesaki flow as intrinsic time; OPH’s Einstein branch consumes modular structure the same way but insists on a finite, producer-typed route to the premises. Holographic and tensor-network programs relate bulk geometry to boundary entanglement; OPH’s screen is a boundary object throughout, and its capacity identities are finite-algebra statements at finite cutoff. Thermodynamic derivations of the Einstein equation from entropy stationarity are the closest relatives of Theorem 6.1; the program’s contribution there is the typed decomposition into separately certifiable premises with executable instruments. Distributed-computation views of physical consensus, quantum error correction, and relational interpretations each anticipate one ingredient; none, to our knowledge, combines finite observer consistency, confluent repair with formal certificates, measured signature instruments with mechanism controls, and an exact finite route to the Standard Model coefficient algebra and global form.

The novelty claim is therefore compositional. No single ingredient is unprecedented; the maintained chain from three observer axioms to the labeled structures above, under a verification and falsification discipline that a hostile auditor can execute, is the contribution offered for evaluation.

Two methodological positions are worth stating for a philosophy-of-physics readership. First, the program treats *evidence-class separation* as part of the theory’s content: a reconstruction whose

conditional joints are hidden is a different, weaker object than the same mathematics with its joints displayed, and machine-enforced receipts require the display. Second, the program treats *observer consistency as prior to state*: A3 selects states from agreement constraints, reversing the usual order in which states are fundamental and intersubjective agreement is emergent. Whether that reversal ultimately earns the physics is exactly what the open gates of Section 12 will decide.

What this paper does not claim

For maximal clarity: no completed prediction of the fine-structure constant; no derivation of the particle mass spectrum; no completed cosmology or dark-sector claim; no claim that the simulator outputs our universe; no claim that formal verification establishes physical truth; no probabilistic certainty language. Downstream research programs exploring such directions exist within the project and are deliberately excluded from the case made here.

Reproducibility statement

All theorems, certificates, simulation configurations, receipts, and the formal library are public in the project repositories [2, 3], pinned by content hash and rebuildable from clean checkouts. The claim ledgers and the frozen prediction register are machine readable [4]. The paper stack from which this synthesis draws, including the component papers on consensus, spacetime and Einstein dynamics, gauge structure, screen microphysics, the particle surfaces, the fixed-point certificates, the Koide identity, and the de Sitter capacity mechanism, is available at the same location [1].

Author contributions and declarations

Author roles, correspondence, data availability, code availability, ethics, competing interests, and AI-assistance declarations follow the synthesis paper of the project stack [1] and apply to this condensation unchanged. AI-assisted tools were used for language editing, drafting assistance, and formatting; the authors reviewed and are responsible for all scientific content, claims, references, and declarations.

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